

Mathematical Foundations of Political Methodology

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Constrained Optimization

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- Method of Lagrange Multipliers with equality constraints
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Optimizing with Constraint

- Problem: maximize (or minimize) $f(x, y)$ subject to constraint that $g(x, y) = 0$
- With two unknowns, possible to solve for $g(x, y)$ in terms of y , then plug in.

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- So now problem is:

$$\max U\left(x, \frac{5x - W}{-4}\right) = x^{\frac{1}{3}} \left(\frac{5x - W}{-4}\right)^{\frac{2}{3}}$$

$$\frac{d}{dx} x^{\frac{1}{3}} \left(\frac{5x - W}{-4} \right)^{\frac{2}{3}} = 0$$

$$\begin{aligned} \frac{d}{dx} x^{\frac{1}{3}} \left(\frac{5x - W}{-4} \right)^{\frac{2}{3}} &= 0 \\ = \frac{1}{3} x^{-\frac{2}{3}} \left(\frac{5x - W}{-4} \right)^{\frac{2}{3}} + x^{\frac{1}{3}} \frac{2}{3} \left(\frac{5x - W}{-4} \right)^{\frac{-1}{3}} \left(\frac{-5}{4} \right) &= 0 \end{aligned}$$

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& (5x - W) = -10x \\
& x = \frac{W}{15}
\end{aligned}$$

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$$\frac{\partial \mathcal{L}}{\partial x} = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = 0, \quad \frac{\partial \mathcal{L}}{\partial \lambda} = 0$$

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- Step 2, plug in:

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$$= xy - \lambda(5x + 2y - 10)$$

- Step 3, take partials to find $\nabla \mathcal{L}$,

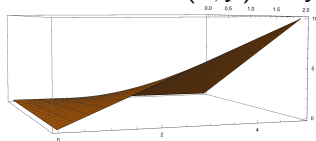
$$\nabla \mathcal{L} = \begin{bmatrix} y - 5\lambda \\ x - 2\lambda \\ -5x - 2y + 10 \end{bmatrix} = \mathbf{0}$$

- Solve for critical points, x , y , λ .

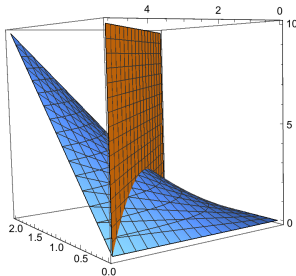
Plot to confirm

- We saw that $x = 1$, $y = \frac{2}{5}$ is the critical point, is it a max?

The function $f(x, y) = xy$



The function $f(x, y) = xy$ and the constraint $y = \frac{10-5x}{2}$



The method of Lagrange multipliers will find all constrained maxima *and* minima, so you need to be careful to check your solutions

(Question: Why didn't our last example find a constrained *minimum*?)

Example: Maximize $5x + y$ subject to $x^2 + y^2 = 10$

$$L(x, y, \lambda) = 5x + y - \lambda(x^2 + y^2 - 10)$$

$$\nabla L = \begin{bmatrix} 5 - 2\lambda x \\ 1 - 2\lambda y \\ -x^2 - y^2 + 10 \end{bmatrix} = \mathbf{0}$$

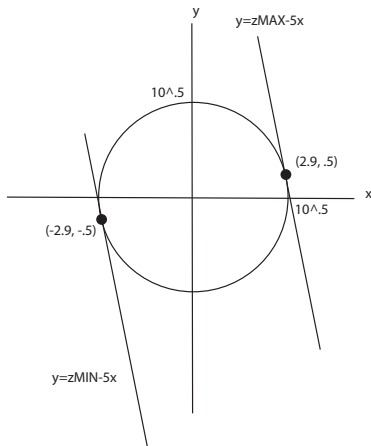
Solving for x, y, λ : $x = \frac{5}{2\lambda}$, $y = \frac{1}{2\lambda}$,

$$\frac{25}{4\lambda^2} + \frac{1}{4\lambda^2} - 10 = 0$$

and so $\lambda = \pm 0.851$ and $(x, y) = (2.93, 0.58)$ and $(-2.93, -0.58)$

Maximize $5x + y$ subject to $x^2 + y^2 = 10$

$(x, y) = (2.93, 0.58)$ is our max $(-2.93, -0.58)$ is our min



Warnings

- If your constraint isn't really a constraint, L won't find it
- Method provides a *necessary* but not *sufficient* condition for optima
- A constrained optimum will be a critical point of the Lagrangian, but *not necessarily a local maximum of the Lagrangian* (i.e. it could be a saddle of L)
- With one constraint, the gradient of the constraint at an optimum can't equal $\mathbf{0}$

What is λ ?

Consider a slightly different problem where we maximize $f(x, y)$ subject to $g(x, y) = b$, with b a constant

The maximum value that f can attain will generally depend on b ; call this maximum value $v(b)$

If (x^*, y^*) is a constrained maximum of f with corresponding λ , then $\lambda = v'(b)$

In words, λ is the rate of change of the quantity being optimized as a function of the constraint parameter b . Economically, it is how much you would pay to loosen the budget constraint.

More variables and more constraints

If we have more than one constraint we create our Lagrangian with a different multiplier for each constraint (though we can't have more constraints than variables)

If we have more than two variables we simply take more first-order conditions to find our critical points

- Potential problem: It's harder to characterize our critical points because we can't draw them out!

Concavity and Maxima

Suppose that $f(\mathbf{x})$ is a concave function, that $g(\mathbf{x}), h(\mathbf{x}), \dots$ are linear, and that we want to maximize f subject to $g, h, \dots$

Then if $L = f - \lambda(g) - \mu(h) - \dots$ has a critical point at $\mathbf{x}^*$, f attains a constrained (global) maximum at $\mathbf{x}^*$

This extends in the obvious way to convex f and global minima

$\Rightarrow$ Convexity or concavity of the objective function along with linear constraints yields unambiguous critical points

A final example:

Maximize $-x^2 - 2y^2 - 3z^2$ subject to $x + y = 5$ and $2x + y + z = 10$

$$L(x, y, z, \lambda, \mu) = -x^2 - 2y^2 - 3z^2 - \lambda(x + y - 5) - \mu(2x + y + z - 10)$$

$$\nabla L = \begin{bmatrix} -2x - \lambda - 2\mu \\ -4y - \lambda - \mu \\ -6z - \mu \\ -x - y + 5 \\ -2x - y - z + 10 \end{bmatrix} = \mathbf{0}$$

$$\nabla L = \begin{bmatrix} -2x - \lambda - 2\mu \\ -4y - \lambda - \mu \\ -6z - \mu \\ -x - y + 5 \\ -2x - y - z + 10 \end{bmatrix} = \mathbf{0}$$

Solving (5 linear equations in 5 unknowns; I used R) we get

$$x = \frac{25}{6}, y = \frac{5}{6}, z = \frac{5}{6}, \lambda = \frac{5}{3}, \mu = -5$$

Since our objective function is concave and our constraints are linear we know that $x^* = \frac{25}{6}, y^* = \frac{5}{6}, z^* = \frac{5}{6}$ is a constrained optimum

$$\nabla L = \begin{bmatrix} -2x - \lambda - 2\mu \\ -4y - \lambda - \mu \\ -6z - \mu \\ -x - y + 5 \\ -2x - y - z + 10 \end{bmatrix} = \mathbf{0}$$

```
> xmat<-matrix(c(-2,0,0,-1,-2,
+ 0,-4,0,-1,-1,
+ 0,0,-6,0,-1,
+ -1,-1,0,0,0,
+ -2,-1,-1,0,0), nrow=5, byrow=TRUE)
>
> solve(xmat)%*%c(0,0,0,-5,-10)
      [,1]
[1,]  4.166667
[2,]  0.8333333
[3,]  0.8333333
[4,]  1.666667
[5,] -5.0000000
>
```